

BASE-FACTOR DESCENT AND EVEN-BASE AUTOMATIC RIGIDITY FOR COLLATZ-INVARIANT SEQUENCES

MUTEKI

ABSTRACT. Let $a = (a_n)_{n \geq 0}$ be a sequence over a finite alphabet and let

$$T(2n) = n, \quad T(2n+1) = 3n+2$$

be the shortcut Collatz map. We prove that for every even base $B \geq 2$, every B -automatic sequence satisfying $a_{T(n)} = a_n$ is constant on the positive integers. The structural input is a base-factor descent principle: if $a_{pn} = a_n$, then

$$\mathcal{K}_b(a) \subseteq \mathcal{K}_{pb}(a),$$

so (pb) -automaticity implies b -automaticity. For an even base $B = 2^s m$ with m odd, shortcut-Collatz invariance therefore descends automaticity from B to m . If $m > 1$, the bases B and m are multiplicatively independent, and Cobham's theorem yields ultimate periodicity. A base-independent argument then collapses every eventual period by means of the identities $b_{4n} = b_{3n}$ and $b_{4n+2} = b_n$ for $b_n = a_{2n+1}$. If $m = 1$, descent reduces to base 2, where the ARC2 theorem applies. We also provide a Lean 4 / Mathlib formalization. In the formal artifact Cobham's theorem is supplied as an explicit theorem parameter rather than reproved internally; all remaining proof obligations, including an independent re-derivation audit of the final theorem, were checked without `sorryAx`.

1. INTRODUCTION

Define the shortcut Collatz map $T : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ by

$$T(2n) = n, \quad T(2n+1) = 3n+2. \quad (1)$$

A finite-valued sequence $a : \mathbb{N}_0 \rightarrow \Delta$ is called *T-invariant* if

$$a_{T(n)} = a_n \quad (n \geq 0). \quad (2)$$

Equivalently,

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2}. \quad (3)$$

The purpose of this paper is to isolate a cross-base mechanism implicit in the first identity and to use it to treat every even automatic base at once.

Date: 19 September 2026.

2020 Mathematics Subject Classification. Primary 11B85; Secondary 11B83, 68Q45.

Key words and phrases. Collatz conjecture, automatic sequence, finite kernel, Cobham theorem, base-factor descent, invariant coloring, Lean 4.

Theorem 1.1 (Main theorem). *Let $B \geq 2$ be even, let Δ be finite, and let $a : \mathbb{N}_0 \rightarrow \Delta$ be B -automatic. If a is T -invariant, then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n > 0).$$

In general no conclusion is possible for a_0 .

The proof has two branches. Write $B = 2^s m$ with $s > 0$ and m odd. The identity $a_{2n} = a_n$ allows automaticity to descend through powers of 2. If $m > 1$, then a is both B -automatic and m -automatic; since an even base and an odd base are multiplicatively independent, Cobham's theorem [6] gives ultimate periodicity. We then prove that every ultimately periodic T -invariant finite coloring is constant on $\mathbb{N}_{>0}$. If $m = 1$, then B is a pure power of 2, and the same descent reduces the problem to base 2.

The result does not resolve the Collatz conjecture. It is a rigidity theorem under the additional finite-state hypothesis of automaticity. The formulation is motivated by earlier ARC cases in bases 2, 3, 5, and 7, but the argument below is organized so that the even-base theorem is mathematically self-contained apart from the previously established base-2 endpoint [9] and the standard Cobham theorem.

2. AUTOMATIC SEQUENCES AND KERNELS

For $k \geq 2$, the k -kernel of a sequence $a : \mathbb{N}_0 \rightarrow \Delta$ is

$$\mathcal{K}_k(a) = \left\{ (a_{k^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < k^e \right\}.$$

We use the standard kernel characterization of automatic sequences; see, for example, [1, Ch. 6].

Theorem 2.1 (Eilenberg kernel characterization). *Let $k \geq 2$ and let $a : \mathbb{N}_0 \rightarrow \Delta$ take values in a finite set. Then a is k -automatic if and only if $\mathcal{K}_k(a)$ is finite.*

3. BASE-FACTOR DESCENT

Lemma 3.1 (Iterated scaling invariance). *Let $p \geq 1$. If $a_{pn} = a_n$ for all $n \geq 0$, then*

$$a_{p^e n} = a_n$$

for all $e, n \geq 0$.

Proof. Induct on e . □

Theorem 3.2 (Base-factor descent). *Let $p \geq 1$ and $b \geq 2$, and suppose $a : \mathbb{N}_0 \rightarrow \Delta$ satisfies $a_{pn} = a_n$ for every $n \geq 0$. Then*

$$\mathcal{K}_b(a) \subseteq \mathcal{K}_{pb}(a). \tag{4}$$

Consequently, if a is (pb) -automatic, then a is b -automatic.

Proof. Take $u = (u_n)_{n \geq 0} \in \mathcal{K}_b(a)$, say

$$u_n = a_{b^e n + r}, \quad 0 \leq r < b^e.$$

By Lemma 3.1,

$$u_n = a_{p^e(b^e n + r)} = a_{(pb)^e n + p^e r}.$$

Since $p^e r < (pb)^e$, we have $u \in \mathcal{K}_{pb}(a)$. The automaticity statement follows from Theorem 2.1. $\square$

For a T -invariant sequence, (3) gives $a_{2n} = a_n$. Hence Theorem 3.2 yields

$$(2b)\text{-automatic} \implies b\text{-automatic}. \quad (5)$$

Iterating, or applying the theorem directly with $p = 2^s$, gives

$$(2^s b)\text{-automatic} \implies b\text{-automatic}. \quad (6)$$

Remark 3.3. The inclusion (4) is elementary. The point is not the difficulty of the lemma but its role as a reusable transfer principle: the dilation invariance supplied by the even Collatz branch converts finite-state information in an even base into finite-state information in its odd part. Related themes appear in the literature on decimation invariance and recognizable sets under affine maps [2, 8, 4]; we do not claim priority for every elementary ingredient in isolation.

4. EVEN BASES AND THE ODD PART

Let $B \geq 2$ be even. Write

$$B = 2^s m, \quad s > 0, \quad m \text{ odd}. \quad (7)$$

Such a factorization exists by repeatedly dividing by 2.

Proposition 4.1 (Descent to the odd part). *Let a be T -invariant and B -automatic, where (7) holds. Then a is m -automatic.*

Proof. Apply (6) with $b = m$. $\square$

Lemma 4.2 (Multiplicative independence). *If $m > 1$ in (7), then B and m are multiplicatively independent.*

Proof. If $B^u = m^v$ for positive integers u, v , then B^u is even whereas m^v is odd, a contradiction. $\square$

Combining Proposition 4.1, Lemma 4.2, and Cobham's theorem gives the following.

Proposition 4.3 (Ultimate periodicity in the non-pure branch). *Assume $m > 1$ in (7). If a is T -invariant and B -automatic, then a is ultimately periodic.*

Proof. The sequence is both B -automatic and m -automatic, and the bases are multiplicatively independent. Cobham's theorem [6] therefore implies ultimate periodicity. $\square$

5. A BASE-INDEPENDENT PERIOD-COLLAPSE THEOREM

The remaining argument no longer depends on an automatic base. Define

$$b_n := a_{2n+1}.$$

Lemma 5.1 (Two identities). *If a is T -invariant, then for all $n \geq 0$,*

$$b_{4n} = b_{3n}, \quad b_{4n+2} = b_n. \quad (8)$$

Proof. Using (3),

$$b_{4n} = a_{8n+1} = a_{12n+2} = a_{6n+1} = b_{3n},$$

and

$$b_{4n+2} = a_{8n+5} = a_{12n+8} = a_{6n+4} = a_{3n+2} = b_n.$$

□

Proposition 5.2 (Collapse of an eventual period). *Suppose $b : \mathbb{N}_0 \rightarrow \Delta$ is ultimately periodic and satisfies (8). Then b is eventually constant.*

Proof. Choose the least positive period P of a periodic tail of b . Passing to residue classes modulo P on a sufficiently far tail gives a map

$$f : \mathbb{Z}/P\mathbb{Z} \rightarrow \Delta$$

satisfying

$$f(4x) = f(3x), \quad f(4x+2) = f(x). \quad (9)$$

We show that $P = 1$.

First, P is odd. Put $d = P/\gcd(P, 4)$. If $\gcd(P, 4) > 1$, then $0 < d < P$ and $4d \equiv 0 \pmod{P}$. From the second identity in (9),

$$f(x+d) = f(4(x+d)+2) = f(4x+2) = f(x),$$

so d is a smaller period, contradicting minimality. Thus $\gcd(P, 4) = 1$.

Next, $3 \nmid P$. If $P = 3d$, then $4d \equiv d \pmod{P}$ and $3d \equiv 0 \pmod{P}$. Applying the first identity to $x+d$ gives

$$f(4x+d) = f(3x).$$

Since also $f(4x) = f(3x)$, we obtain $f(4x+d) = f(4x)$. Because 4 is invertible modulo P , this implies $f(y+d) = f(y)$ for every y , again contradicting minimality. Hence $\gcd(P, 12) = 1$.

Let $q = 4^{-1}$ in $\mathbb{Z}/P\mathbb{Z}$ and define the bijections

$$S(x) = 3qx, \quad F(x) = 4x + 2.$$

The two identities in (9) imply

$$f(Sx) = f(x), \quad f(Fx) = f(x).$$

Since S and F are bijections, f is also invariant under their inverses. A direct calculation gives

$$S \circ F \circ S^{-1} \circ F^{-1}(x) = x - 2^{-1}.$$

Therefore $f(x - 2^{-1}) = f(x)$ for every x . Because P is odd, 2^{-1} is a unit modulo P , so translation by -2^{-1} is transitive on $\mathbb{Z}/P\mathbb{Z}$. Hence f is constant, forcing $P = 1$ by the minimality of P . $\square$

Lemma 5.3 (Propagation from the tail). *If b is eventually constant and satisfies $b_{4n+2} = b_n$, then b is constant on all of $\mathbb{N}_0$.*

Proof. Let $F(n) = 4n + 2$. Then $b_{F(n)} = b_n$, and

$$F^k(n) = 4^k n + \frac{2(4^k - 1)}{3} \rightarrow \infty.$$

For fixed n , choose k large enough that $F^k(n)$ lies in the constant tail. Then $b_n = b_{F^k(n)}$ equals the tail value. $\square$

Theorem 5.4 (Base-independent positive rigidity). *If $a : \mathbb{N}_0 \rightarrow \Delta$ is T -invariant and ultimately periodic, then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n > 0).$$

Proof. The subsequence $b_n = a_{2n+1}$ is ultimately periodic. By Lemma 5.1, Proposition 5.2, and Lemma 5.3, there is $c \in \Delta$ with $a_{2n+1} = c$ for every $n \geq 0$. Every positive integer can be written as $2^s(2m+1)$. Repeatedly applying $a_{2n} = a_n$ gives

$$a_{2^s(2m+1)} = a_{2m+1} = c.$$

$\square$

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Write $B = 2^s m$ with $s > 0$ and m odd.

If $m > 1$, Proposition 4.3 gives ultimate periodicity, and Theorem 5.4 gives constancy on the positive integers.

If $m = 1$, then $B = 2^s$. Since $s > 0$, write $s = t + 1$. By Theorem 3.2, base- 2^{t+1} automaticity descends to base 2. The conclusion then follows from the base-2 ARC theorem [9]. $\square$

Corollary 6.1. *Every even base $B \geq 2$ is covered by Theorem 1.1. In particular the descents*

$$6 \rightarrow 3, \quad 10 \rightarrow 5, \quad 14 \rightarrow 7$$

are immediate examples of the base-factor mechanism.

7. FORMALIZATION IN LEAN 4

The proof was formalized in Lean 4 / Mathlib in a modular development. The principal endpoints are summarized below. The formal artifact uses finite-kernel automaticity as its main interface and bridges to explicit MSD automata when invoking the Cobham parameter.

Stage	Mathematical role
-------	-------------------

1	General $pb \rightarrow b$ kernel descent.
2–3	Extraction of the shortcut even/odd branches and descent through arbitrary powers 2^s .
4–5	Packaging of power-of-two towers, descent to the odd part, and multiplicative independence.
6	Conversion to two MSD-automatic descriptions and the Cobham handoff to ultimate periodicity.
7	Base-independent collapse of eventual periods and positive-domain rigidity.
8	Pure power-of-two branch via ARC2 and integration of the two branches.
9	Internal construction of $B = 2^s m$ and final closure for every even $B \geq 2$.

The principal Lean endpoints are `arc_baseFactor_automatic_descent`, `arc_powTwoFactor_automatic_descent`, `arc_evenBase_ultimatelyPeriodic_of_cobham`, `arc_ultimatelyPeriodic_shortcut_positive_rigidity`, and the final theorem `arc_evenBase_final_of_cobham`.

The final theorem in the Lean artifact is parameterized by `ARCCobhamPrinciple`, which represents the standard Cobham theorem at the point where the two automatic bases are known to be multiplicatively independent. Cobham’s theorem itself is not reproved in the present Lean development. Thus the mechanized statement is best read as a formal verification of the reduction from the ARC hypotheses to Cobham plus the remaining period-collapse argument.

As an additional audit, `ARCBASEFactorDescentFinalAudit.lean` independently re-derives the all-even theorem from lower-level components without invoking the packaged Stage 8/9 final wrappers. It reconstructs the pure-power branch through Stage 3 and ARC2, reconstructs the non-pure branch through Stages 6–7, and includes concrete smoke tests in bases 4, 18, 64, and 100.

For the Stage 9 endpoint and the independent audit, `#print axioms` reports no `sorryAx`; the remaining logical dependencies are the standard Lean/Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`. Because Cobham is supplied as an ordinary theorem parameter rather than a registered global axiom, it does not appear in this axiom list; this should not be interpreted as an internal formal proof of Cobham.

8. SHARPNESS AT ZERO

The exclusion of zero in Theorem 1.1 is necessary. Given arbitrary $x, y \in \Delta$, define

$$a_0 = x, \quad a_n = y \quad (n \geq 1).$$

Then a is automatic in every base and satisfies (3). Hence a_0 is genuinely unconstrained.

9. RELATION TO PREVIOUS WORK

The kernel characterization of automatic sequences and the basic closure theory are standard [1]. Cobham’s theorem is due to Cobham [6]; see also the survey [3]. Decimation-invariant sequences and related automaticity questions were studied by Barbe and Skordev [2], while recognizable sets under affine maps were investigated by Klarner [8] and Boigelot [4]. Collatz dynamics have also been studied through finite automata, transducers, cellular automata, and base conversion; examples include [10, 7, 11, 5].

The present theorem concerns a different object: a one-variable finite coloring invariant along every shortcut-Collatz edge. In the literature surveyed for this work, we did not locate a theorem stated in precisely the combination used here: the cross-base kernel inclusion induced by $a_{pn} = a_n$, its specialization to shortcut-Collatz invariant colorings, and its integration with Cobham and the common period-collapse argument to cover every even base. Since the kernel inclusion itself is elementary, an equivalent formulation may well occur elsewhere. The contribution claimed here is the structural organization of these ingredients into a single even-base rigidity theorem rather than priority over each component separately.

10. CONCLUDING REMARKS

Theorem 1.1 reduces the even-base part of the ARC program to a single structural mechanism. The remaining generalization problem is therefore concentrated in odd bases. Composite odd bases such as 15 provide natural stress tests for identifying which parts of the base-5 and base-7 finite-state arguments extend uniformly.

DECLARATIONS

Formalization availability. The Lean 4 source files and the independent audit file are intended to accompany this manuscript as supplementary material or in the associated public repository.

Use of generative AI. OpenAI ChatGPT was used as an assistance tool for exploration, Lean code drafting and revision, literature-search support, adversarial checking, and manuscript drafting. The AI system is not an author or a primary source. Responsibility for the mathematical claims, citations, formalization results, and submitted text remains with the human author.

REFERENCES

- [1] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, Cambridge, 2003.
- [2] A. Barbe and G. Skordev, Decimation-invariant sequences and their automaticity, *Theoret. Comput. Sci.* **259** (2001), 379–403.
- [3] J. P. Bell, Cobham’s theorem and its extensions, *Actes des rencontres du CIRM* **1** (2009), no. 1, 11–16.

- [4] B. Boigelot, On iterating linear transformations over recognizable sets of integers, *Theoret. Comput. Sci.* **309** (2003), 413–468.
- [5] D. Caucal and C. Rispal, On the powers of the Collatz function, in *Machines, Computations, and Universality (MCU 2024), Revised Selected Papers*, Lecture Notes in Comput. Sci., Springer, 2025, pp. 115–129.
- [6] A. Cobham, On the base-dependence of sets of numbers recognizable by finite automata, *Math. Systems Theory* **3** (1969), 186–192, doi:10.1007/BF01746527.
- [7] J. Kari, Cellular automata, the Collatz conjecture and powers of $3/2$, in *Developments in Language Theory (DLT 2012)*, Lecture Notes in Comput. Sci. **7410**, Springer, 2012, pp. 40–49.
- [8] D. A. Klarner, m -recognizability of sets closed under certain affine functions, *Discrete Appl. Math.* **21** (1988), no. 3, 207–214.
- [9] Muteki, Automatic rigidity for Collatz-invariant 2-automatic sequences, preprint, 2026.
- [10] J. Shallit and D. A. Wilson, The “ $3x + 1$ ” problem and finite automata, *Bull. Eur. Assoc. Theor. Comput. Sci.* **46** (1992), 182–185.
- [11] T. Sterin and D. Woods, The Collatz process embeds a base conversion algorithm, in *Reachability Problems (RP 2020)*, Lecture Notes in Comput. Sci. **12448**, Springer, 2020, pp. 131–147.

INDEPENDENT RESEARCHER, JAPAN